Sindh Science Learning Portal

Class 5 · Science

Chapter 7

Light and Sound



Learn about sources of light, how shadows form, transparent and opaque things, and how sound is made and travels.

Following the Sindh Textbook Board curriculum

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CHAPTER 7

Light and Sound

What we will learn Learn about sources of light, how shadows form, transparent and opaque things, and how sound is made and travels.

1 Sources of light

Things that give out their own **light** are called **luminous objects** or **sources** of light. Examples: the sun, a bulb, a candle, a firefly.

The sun is a **natural** source of light. Bulbs and torches are **man-made** sources.

The moon, a book and a chair do not give out their own light. They are **non-luminous**. The moon reflects sunlight, so it looks bright.

We see an object when light from it enters our **eyes**.



Picture: Kelvinsong · CC0 · Wikimedia Commons

2 Light and shadows

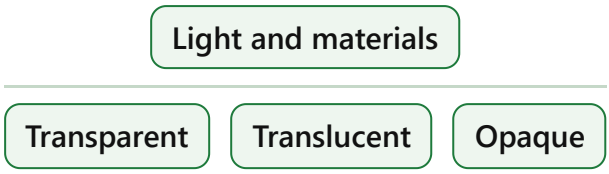
Light travels in **straight lines**.

When an object blocks the path of light, a dark area forms behind it. This is called a **shadow**.

Materials are of three kinds:

- **Transparent**: all the light passes through and we can see clearly. Examples: clear glass, clean water.
- **Translucent**: some light passes through and things look blurry. Examples: butter paper, frosted glass.
- **Opaque**: no light passes through and a shadow forms. Examples: wood, a book, a wall.

Diagram: the parts and what they do



Light and materials (روشني ۽ شيون · روشنی اور اشیاء)
Light passes through some materials but not others.

Transparent (شفاف · شفاف)
All light passes through. Examples: clear glass, clean water.

Translucent (نيم شفاف · نیم شفاف)
Some light passes through; things look blurry. Example: butter paper.

Opaque (غير شفاف · غیر شفاف)
No light passes through and a shadow forms. Examples: wood, a book.

3 Sound

Sound is made when something **vibrates** (moves quickly back and forth). Beat a drum or pluck a stretched rubber band and you can feel the vibration.

Sound can travel through **solids, liquids and gases**. It passes through air, water and even walls.

Sound cannot travel through **empty space**, because there is no matter there.

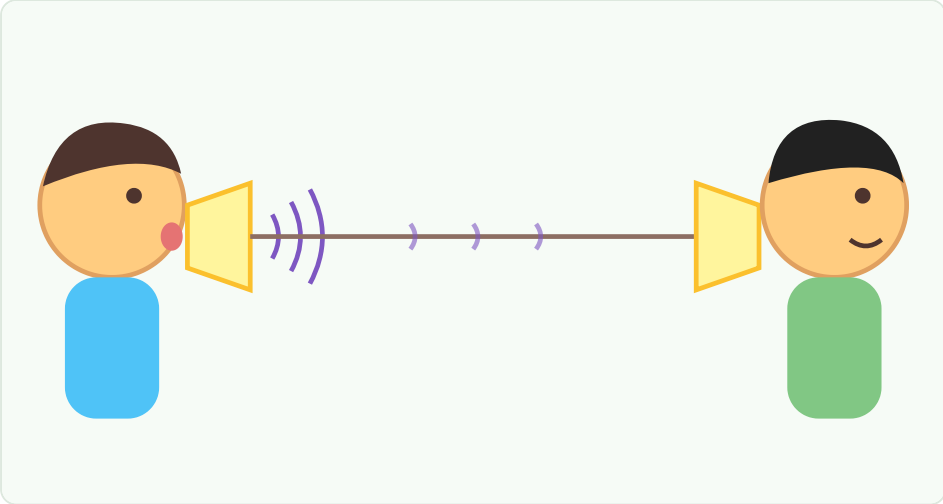
Some sounds are **loud** and some are **soft**. Very loud sounds can harm our ears, so we should protect our ears.

Key words

English	سنڌي	اردو
light	روشني	روشنی
luminous object	روشن شيءِ	روشن جسم
non-luminous object	غير روشن شيءِ	غیر روشن جسم
shadow	پاڇو	سایہ
transparent	شفاف	شفاف
translucent	نيم شفاف	نیم شفاف
opaque	غير شفاف	غیر شفاف
sound	آواز	آواز
vibration	ٿرٿراهٽ	ارتعاش

English	سنڌي	اردو
space (vacuum)	خلا	خلا

Experiment: Make a string telephone



You need

- Two paper or plastic cups
- About 5 metres of strong string
- A pencil (to make holes)

Steps

1. With an adult's help, make a small hole in the bottom of each cup with the pencil.
2. Push each end of the string through a hole and tie a knot inside the cup.
3. Stand far from a friend so the string is tight.
4. Speak softly into one cup while your friend listens with the other cup to their ear.

Safety: Ask an adult to help make the holes, and never shout into the cup.

What did we learn? Your friend can hear you clearly. Your voice makes the cup and the tight string vibrate, and the vibrations travel along the string (a solid) to the other cup. This shows that sound travels through solids.

Practice questions

Circle the correct answer

1. **Which of these gives out its own light?**

   (a) The moon
   (b) The sun
   (c) A mirror
   (d) A book

2. **Why does the moon look bright at night?**

(a) It reflects sunlight (b) It is on fire

(c) It has bulbs on it (d) It is a star

3. **When can we see an object?**

(a) When light from it enters our eyes (b) Only when it makes a loud sound

(c) When our eyes are closed (d) Only in the dark

4. **How does light travel?**

(a) In straight lines (b) In circles

(c) In wavy curves only (d) It does not travel

5. **When does a shadow form?**

(a) When an opaque object blocks the light (b) When there is no light at all

(c) When the object is transparent (d) When there is a sound

6. **Clear glass is which kind of material?**

(a) Transparent (b) Translucent

(c) Opaque (d) Luminous

7. **How is sound made?**

(a) By vibrations (b) By light

(c) By shadows (d) By cold

8. **Where can sound NOT travel?**

(a) In air (b) In water

(c) In empty space (d) Through a wall

9. **Which of these is a man-made source of light?**

(a) The sun (b) A firefly

(c) A torch (d) A star

Write true or false

10. **The moon gives out its own light.** _____

11. **Wood is opaque, so it makes a shadow.** _____

12. **Butter paper is translucent.** _____

13. **Sound cannot travel through water.** _____

14. **Very loud sounds can harm our ears.** _____

Match the columns

15. **Match each material with its kind.**

1. Clean water		(a) Translucent
2. Frosted glass		(b) Opaque
3. Wall		(c) Transparent

Match the columns

16. **Match each word with its meaning.**

1. Luminous object		(a) Dark area behind blocked light
2. Shadow		(b) Quick back-and-forth movement
3. Vibration		(c) Gives its own light

Worksheet: Light and Sound

Name: _____ Date: _____

A. Fill in the blanks

Word box: luminous · shadow · space · straight · vibrations

1. Things that give out their own light are called _____ objects.
2. Light travels in _____ lines.
3. A _____ forms behind an opaque object.
4. Sound is made by _____ .
5. Sound cannot travel through empty _____ .

B. Put ✓ for true and ✗ for false

1. ☐ The sun is a natural source of light.
2. ☐ Light cannot pass through clear glass.
3. ☐ Sound travels through air.
4. ☐ A book is a luminous object.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Candle	( ) Sound from vibration
2. Chair	( ) Translucent
3. Drum	( ) Luminous object
4. Butter paper	( ) Non-luminous object

D. Answer in short

1. Write three sources of light.

2. How is a shadow formed?

3. Through which three kinds of matter can sound travel?

E. Draw and label

Draw the sun, a tree and the tree's shadow. Show the path of light with arrows.

FOR TEACHERS

Answer key

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Practice questions

1. (b) The sun
2. (a) It reflects sunlight
3. (a) When light from it enters our eyes
4. (a) In straight lines
5. (a) When an opaque object blocks the light
6. (a) Transparent
7. (a) By vibrations
8. (c) In empty space
9. (c) A torch
10. False
11. True
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (c), 2 → (a), 3 → (b)

Worksheet

A. Fill in the blanks: 1. luminous · 2. straight · 3. shadow · 4. vibrations · 5. space

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Sound from vibration · (4) Translucent · (1) Luminous object · (2) Non-luminous object

D. Answer in short:

1. The sun, a bulb and a candle.
2. When an opaque object blocks light, a dark area forms behind it.
3. Solids, liquids and gases.

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